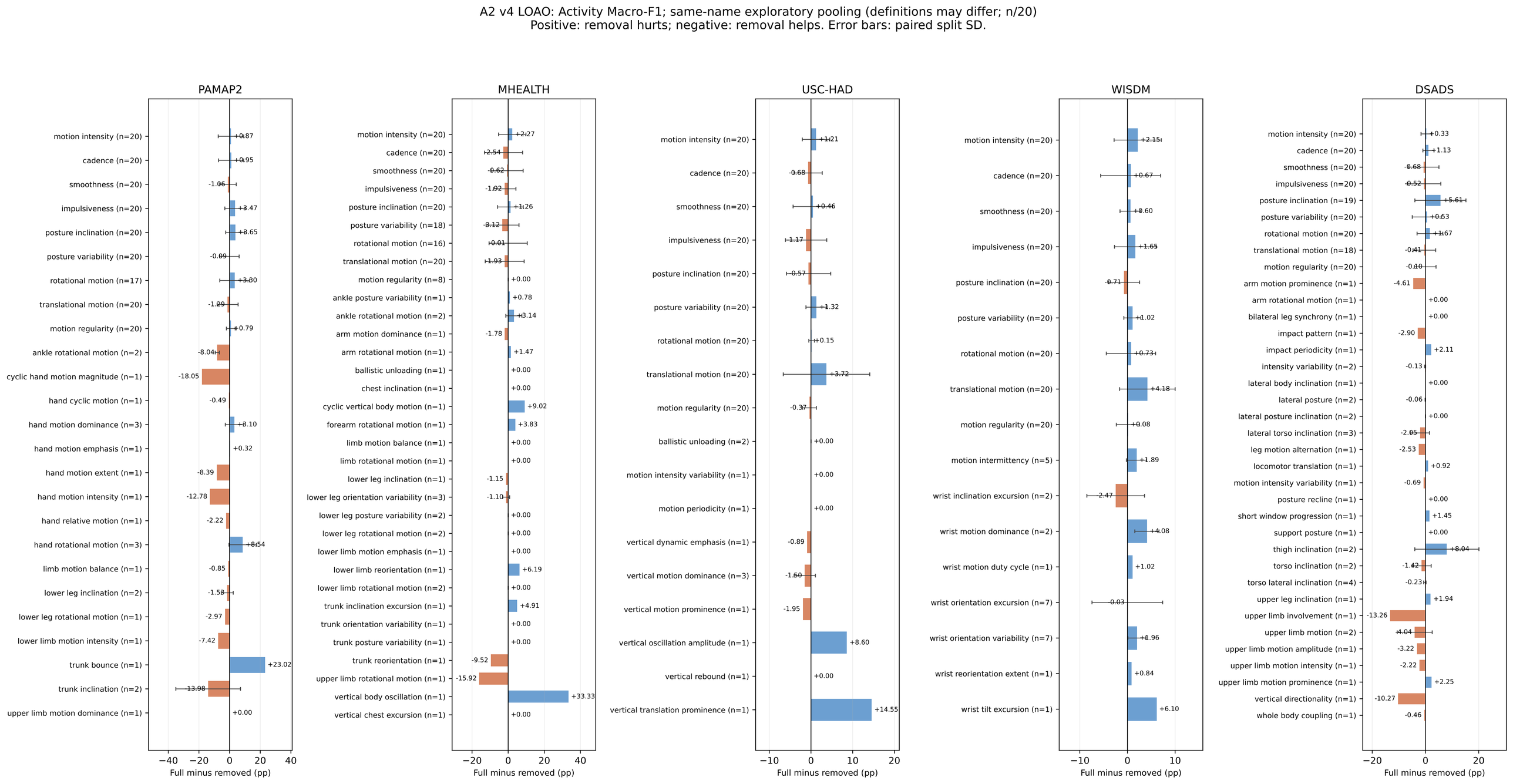
A2 v4 LOAO: Activity Macro-F1; same-name exploratory pooling (definitions may differ; n/20)
Positive: removal hurts; negative: removal helps. Error bars: paired split SD.
PAMAP2
motion intensity (n=20) +0.87
cadence (n=20) +0.95
smoothness (n=20) -1.06
impulsiveness (n=20) +3.47
posture inclination (n=20) +8.65
posture variability (n=20) -0.09
rotational motion (n=17) +3.80
translational motion (n=20) -1.29
motion regularity (n=20) +0.79
ankle rotational motion (n=2) -8.04
cyclic hand motion magnitude (n=1) -18.05
hand cyclic motion (n=1) -0.49
hand motion dominance (n=3) +8.10
hand motion emphasis (n=1) +0.32
hand motion extent (n=1) -8.39
hand motion intensity (n=1) -12.78
hand relative motion (n=1) -2.22
hand rotational motion (n=3) +8.54
limb motion balance (n=1) -0.85
lower leg inclination (n=2) -1.58
lower leg rotational motion (n=1) -2.97
lower limb motion intensity (n=1) -7.42
trunk bounce (n=1) +23.02
trunk inclination (n=2) -13.98
upper limb motion dominance (n=1) +0.00
−40 −20 0 20 40
Full minus removed (pp)
MHEALTH
motion intensity (n=20) +2.27
cadence (n=20) -2.54
smoothness (n=20) -0.62
impulsiveness (n=20) -1.92
posture inclination (n=20) +1.26
posture variability (n=18) -8.12
rotational motion (n=16) +0.01
translational motion (n=20) -1.93
motion regularity (n=8) +0.00
ankle posture variability (n=1) +0.78
ankle rotational motion (n=2) +3.14
arm motion dominance (n=1) -1.78
arm rotational motion (n=1) +1.47
ballistic unloading (n=1) +0.00
chest inclination (n=1) +0.00
cyclic vertical body motion (n=1) +9.02
forearm rotational motion (n=1) +3.83
limb motion balance (n=1) +0.00
limb rotational motion (n=1) +0.00
lower leg inclination (n=1) -1.15
lower leg orientation variability (n=3) -1.10
lower leg posture variability (n=2) +0.00
lower leg rotational motion (n=2) +0.00
lower limb motion emphasis (n=1) +0.00
lower limb reorientation (n=1) +6.19
lower limb rotational motion (n=2) +0.00
trunk inclination excursion (n=1) +4.91
trunk orientation variability (n=1) +0.00
trunk posture variability (n=1) +0.00
trunk reorientation (n=1) -9.52
upper limb rotational motion (n=1) -15.92
vertical body oscillation (n=1) +33.33
vertical chest excursion (n=1) +0.00
−20 0 20 40
Full minus removed (pp)
USC-HAD
motion intensity (n=20) +1.21
cadence (n=20) -0.68
smoothness (n=20) +0.46
impulsiveness (n=20) -1.17
posture inclination (n=20) -0.57
posture variability (n=20) +1.32
rotational motion (n=20) +0.15
translational motion (n=20) +3.72
motion regularity (n=20) -0.37
ballistic unloading (n=2) +0.00
motion intensity variability (n=1) +0.00
motion periodicity (n=1) +0.00
vertical dynamic emphasis (n=1) -0.89
vertical motion dominance (n=3) -1.60
vertical motion prominence (n=1) -1.95
vertical oscillation amplitude (n=1) +8.60
vertical rebound (n=1) +0.00
vertical translation prominence (n=1) +14.55
−10 0 10 20
Full minus removed (pp)
WISDM
motion intensity (n=20) +2.15
cadence (n=20) +0.67
smoothness (n=20) +0.60
impulsiveness (n=20) +1.65
posture inclination (n=20) -0.71
posture variability (n=20) +1.02
rotational motion (n=20) +0.73
translational motion (n=20) +4.18
motion regularity (n=20) +0.08
motion intermittency (n=5) +1.89
wrist inclination excursion (n=2) -2.47
wrist motion dominance (n=2) +4.08
wrist motion duty cycle (n=1) +1.02
wrist orientation excursion (n=7) -0.03
wrist orientation variability (n=7) +1.96
wrist reorientation extent (n=1) +0.84
wrist tilt excursion (n=1) +6.10
−10 0 10
Full minus removed (pp)
DSADS
motion intensity (n=20) +0.33
cadence (n=20) +1.13
smoothness (n=20) -0.68
impulsiveness (n=20) -0.52
posture inclination (n=19) +5.61
posture variability (n=20) +0.63
rotational motion (n=20) +1.67
translational motion (n=18) -0.41
motion regularity (n=20) -0.10
arm motion prominence (n=1) -4.61
arm rotational motion (n=1) +0.00
bilateral leg synchrony (n=1) +0.00
impact pattern (n=1) -2.90
impact periodicity (n=1) +2.11
intensity variability (n=2) -0.13
lateral body inclination (n=1) +0.00
lateral posture (n=2) -0.06
lateral posture inclination (n=2) +0.00
lateral torso inclination (n=3) -2.05
leg motion alternation (n=1) -2.53
locomotor translation (n=1) +0.92
motion intensity variability (n=1) -0.69
posture recline (n=1) +0.00
short window progression (n=1) +1.45
support posture (n=1) +0.00
thigh inclination (n=2) +8.04
torso inclination (n=2) -1.42
torso lateral inclination (n=4) -0.23
upper leg inclination (n=1) +1.94
upper limb involvement (n=1) -13.26
upper limb motion (n=2) -4.04
upper limb motion amplitude (n=1) -3.22
upper limb motion intensity (n=1) -2.22
upper limb motion prominence (n=1) +2.25
vertical directionality (n=1) -10.27
whole body coupling (n=1) -0.46
−20 0 20
Full minus removed (pp)